

Phase I/II Operating Characteristics

Continuous enrollment | 56-day arrival | IPDE alpha = 0.9

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	9.4	4.6	2.3	0.9	0.3	82.5
C2 alpha=0.9 one-stage	8.9	1.8	0.2	0	0	89.1
C3 alpha=0.9 one-stage	9	0.9	0	0	0	90.1
Oracle C1 alpha=0 two-stage	9.2	5.1	2.2	0.8	0.3	82.4
C2 alpha=0.9 two-stage	8.7	2.1	0.2	0	0	89
C3 alpha=0.9 two-stage	9.7	1	0	0	0	89.3
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	11	4.2	2.3	0.9	0.2	
C2 alpha=0.9 one-stage	11.9	2.6	0.8	0.2	0	
C3 alpha=0.9 one-stage	12.5	2	0.5	0	0	
Oracle C1 alpha=0 two-stage	10.9	4.3	2.4	0.9	0.3	
C2 alpha=0.9 two-stage	11.8	2.7	0.9	0.2	0	
C3 alpha=0.9 two-stage	12.5	2.2	0.6	0	0	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	23.9	21	19.3	21.8	13.1	0.9
C2 alpha=0.9 one-stage	23.1	30.8	18.9	17.2	9.8	0.1
C3 alpha=0.9 one-stage	23	31.9	21.7	16	7.5	0
Oracle C1 alpha=0 two-stage	24.1	21.9	18.3	21.1	13.7	0.9
C2 alpha=0.9 two-stage	23.6	32.3	17.3	17.3	9.5	0
C3 alpha=0.9 two-stage	23.9	31.4	24.2	14	6.6	0
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.2	6.7	5.8	5.8	4.5	
C2 alpha=0.9 one-stage	7.2	7.9	5.9	5.3	3.7	
C3 alpha=0.9 one-stage	6.9	8	6.7	5.3	3.1	
Oracle C1 alpha=0 two-stage	7.1	6.7	5.7	5.8	4.8	
C2 alpha=0.9 two-stage	6.8	7.9	5.6	5.6	4.1	
C3 alpha=0.9 two-stage	6.6	7.5	6.7	5.4	3.9	

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	4.9	47	28.4	12.3	2.2	5.2
C2 alpha=0.9 one-stage	6.2	53.9	26.5	6.5	0.4	6.5
C3 alpha=0.9 one-stage	6.6	57.7	25.1	3.2	0.2	7.2
Oracle C1 alpha=0 two-stage	4.5	48.5	29.1	10.9	1.9	5.1
C2 alpha=0.9 two-stage	6.2	56.3	24.7	5.6	0.5	6.8
C3 alpha=0.9 two-stage	7.4	57	25.2	3	0.2	7.2
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	5.2	11.3	7.6	4.1	1.3	
C2 alpha=0.9 one-stage	5.8	12.6	7.4	2.9	0.5	
C3 alpha=0.9 one-stage	5.8	13	7.7	2.3	0.3	
Oracle C1 alpha=0 two-stage	5.1	10.7	8	4.3	1.4	
C2 alpha=0.9 two-stage	5.6	11.9	7.6	3.3	0.8	
C3 alpha=0.9 two-stage	5.6	12.4	7.8	2.8	0.6	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	0.8	1.4	6	59.4	32.4	0
C2 alpha=0.9 one-stage	0.5	0.6	3.3	66.8	28.7	0.1
C3 alpha=0.9 one-stage	0.2	0.7	3.1	65	31.1	0
Oracle C1 alpha=0 two-stage	1.1	1.2	6.3	56	35.4	0
C2 alpha=0.9 two-stage	0.6	0.5	3.7	58.6	36.4	0.1
C3 alpha=0.9 two-stage	0.5	0.5	3.6	53.7	41.7	0
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	3.4	3.5	4.5	12.5	6.2	
C2 alpha=0.9 one-stage	3.4	3.3	4.2	14	5.2	
C3 alpha=0.9 one-stage	3.3	3.3	4.4	13.4	5.5	
Oracle C1 alpha=0 two-stage	3.4	3.4	4.3	10.9	8	
C2 alpha=0.9 two-stage	3.3	3.3	3.9	11.4	8.1	
C3 alpha=0.9 two-stage	3.3	3.3	4	10.7	8.7	

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Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	10.7	25.3	44.6	9.5	0.6	9.3
C2 alpha=0.9 one-stage	13.2	33.1	40.8	2.9	0	9.9
C3 alpha=0.9 one-stage	16.4	36.1	36.2	1.9	0.1	9.3
Oracle C1 alpha=0 two-stage	11.3	23.8	46.3	9	0.9	8.7
C2 alpha=0.9 two-stage	14.5	34.2	38.5	3.7	0	9.1
C3 alpha=0.9 two-stage	16.8	36.6	35	2.2	0	9.4
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.3	8	9.8	3.4	0.6	
C2 alpha=0.9 one-stage	8.4	9.3	9.1	1.9	0.1	
C3 alpha=0.9 one-stage	8.9	10	8.7	1.3	0	
Oracle C1 alpha=0 two-stage	6.9	7.9	9.7	3.8	0.9	
C2 alpha=0.9 two-stage	8	9.3	8.7	2.5	0.4	
C3 alpha=0.9 two-stage	8.6	9.7	8.5	1.9	0.3	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	3	6.5	8.7	11.8	65.3	4.7
C2 alpha=0.9 one-stage	5.2	9.6	12	17.4	51.5	4.4
C3 alpha=0.9 one-stage	5.2	11	17.5	18.8	42.4	5.1
Oracle C1 alpha=0 two-stage	4.6	7.2	8.6	11.6	63.2	4.8
C2 alpha=0.9 two-stage	5.4	11.3	11.9	17.1	49.6	4.7
C3 alpha=0.9 two-stage	6.1	13.1	17.4	17.7	41.1	4.5
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	4.6	4.6	5	5.5	10.2	
C2 alpha=0.9 one-stage	4.8	5.3	5.6	5.7	8.4	
C3 alpha=0.9 one-stage	4.9	5.6	6.2	5.6	7.4	
Oracle C1 alpha=0 two-stage	4.5	4.5	4.9	5.6	10.3	
C2 alpha=0.9 two-stage	4.8	5.2	5.4	5.9	8.6	
C3 alpha=0.9 two-stage	4.8	5.6	6.1	5.6	7.6	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
Oracle C1 alpha=0 one-stage	18.7	30	29.5	30	29.1	29.8	27.8
C2 alpha=0.9 one-stage	10.2	18.1	18.4	19.6	18.7	18.5	17.2
C3 alpha=0.9 one-stage	9.1	15.3	16	17.6	16.4	15.5	15
Oracle C1 alpha=0 two-stage	18.8	30	29.5	30	29.1	29.8	27.9
C2 alpha=0.9 two-stage	10.2	18.2	18.5	19.7	18.9	18.5	17.3
C3 alpha=0.9 two-stage	9.2	15.5	16.2	17.9	16.6	15.6	15.2

Values are reported from Total_Unique_Patients; comparator counts without patient IDs are sums of dose-level means.

Rows are one-stage first, then two-stage. C1 is fixed at alpha=0; C2/C3 use alpha=0.9.